

COURSE TITLE : **WEB APPLICATION DEVELOPMENT**
 COURSE CODE : **4263**
 COURSE CATEGORY : **A**
 PERIODS/WEEK : **5**
 PERIODS/SEMESTER : **75**
 CREDITS : **5**

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Introduction to Web Application and PHP	18
2	Server side scripting using PHP	19
3	Database handling using PHP and MySQL	19
4	Client side scripting using JavaScript	19

Course General Outcomes:

Sl.	G.O	On completion of this course the student will be able to :
1	1	To understand Web Application development using PHP
2	1	To comprehend Server side scripting using PHP
3	1	To understand Database handling using PHP and MySQL
4	1	To understand Client side scripting using JavaScript

Specific Outcomes:

MODULE – I: Introduction to Web Application and PHP

1.1 Understand Web Application development using PHP

- 1.1.1 Define Internet
- 1.1.2 Differentiate Internet and Web
- 1.1.3 Describe Web Servers
- 1.1.4 Describe Working of a LAMP Server
- 1.1.5 Distinguish Dynamic Web Scripting and Web Applications
- 1.1.6 Describe Need for Web Based Applications
- 1.1.7 Explain PHP and its Evolution

- 1.1.8 List Advantages of PHP over other scripting languages
- 1.1.9 List Technologies used in PHP based applications
- 1.1.10 Describe PHP compilers and engines
- 1.1.11 Describe The Zend engine
- 1.1.12 Describe Alternative PHP cache
- 1.1.13 Describe Dual stage execution in PHP

MODULE – II: Server side scripting using PHP

2.1 Comprehend Server side scripting using PHP

- 2.1.1 Explain PHP language Basics
- 2.1.2 Describe PHP syntax and semantics
- 2.1.3 Show Embedding PHP in WebPages
- 2.1.4 Describe PHP variables (concepts and implementation)
- 2.1.5 Discuss Variable scopes
- 2.1.6 Illustrate Data types
- 2.1.7 Describe Importance of \$ symbol
- 2.1.8 Summarise operators and selection statements
- 2.1.9 Explain loops and functions
- 2.1.10 Explain Strings in PHP
- 2.1.11 Illustrate string operations and functions
- 2.1.12 Describe echo and print commands
- 2.1.13 Explain how to define constants
- 2.1.14 Explain PHP Arrays
- 2.1.15 Illustrate Associate arrays and for-each loop
- 2.1.16 Describe how to generate HTML elements dynamically
- 2.1.17 Describe Time management in PHP, UTC, Timestamp
- 2.1.18 Illustrate date function and time function
- 2.1.19 Explain how to generate custom timestamp

MODULE – III: Database handling using PHP and MySQL

3.1 Understand Database handling using PHP and MySQL

- 3.1.1 Describe session
- 3.1.2 Explain inter page communication
- 3.1.3 Explain how session works on LAMP Server
- 3.1.4 List applications of session
- 3.1.5 Discuss session management in PHP and its drawbacks

- 3.1.6 Explain what is Cookies, its needs, working principle
- 3.1.7 Explain Cookie in PHP
- 3.1.8 List Advantages of Cookie over Session
- 3.1.9 Summarize how to manage MySQL database with PHP
- 3.1.10 Show how to establish connection and create Query Strings.
- 3.1.11 Show how to execute Queries & Non-Queries
- 3.1.12 Show how to fetch data from result
- 3.1.13 Explain form Management
- 3.1.14 Distinguish GET and POST forms
- 3.1.15 Discuss how to retrieve data from forms
- 3.1.16 Show how to repopulate HTML forms
- 3.1.17 Explain how to upload files into server and validate uploaded files

MODULE – IV: Client side scripting using javaScript

4.1 Understand Client side scripting using javaScript

- 4.1.1 Define client-side scripting
- 4.1.2 List needs for a client-side language
- 4.1.3 Define javaScript
- 4.1.4 Conclude History and evolution of javaScript
- 4.1.5 Describe Interpreter used for javaScript
- 4.1.6 Illustrate javaScript basics, syntax and semantics
- 4.1.7 Summarise variables, its scope and data types
- 4.1.8 Describe operators and selection statements
- 4.1.9 Explain loops and functions
- 4.1.10 Explain strings and string operations
- 4.1.11 Show embedding javaScript in WebPages
- 4.1.12 Explain arrays in javaScript
- 4.1.13 Discuss HTML events and javaScript functions
- 4.1.14 Describe element selector functions, accessing HTML elements & modifying
- 4.1.15 Show handling html events using event attribute function calls
- 4.1.16 Describe accessing form elements
- 4.1.17 Explain validating html forms

CONTENT DETAILS

MODULE – I: Introduction to Web Application and PHP

Introduction to Internet – Internet and Web differentiation – Introduction to Web Servers – Working of a LAMP Server – Introduction to Dynamic Web Scripting and Web Applications – Need for Web Based Applications – Introduction to PHP – Evolution of PHP - Advantages of PHP over other scripting languages – Technologies used in PHP based applications – PHP compilers and engines – The Zend engine – Alternative PHP cache – Dual stage execution in PHP.

MODULE – II: Server side scripting using PHP

PHP language Basics – PHP syntax and semantics – Embedding PHP in WebPages – PHP variables (concepts and implementation) – Variable scopes - Data types - Importance of '\$' symbol – operators – selection statements – loops – functions – Strings in PHP – string operations and functions – echo and print commands – Defining constants - PHP Arrays – Associate arrays and 'for-each' loop – Generating HTML Elements Dynamically - Time management in PHP – Universal Time Constant (UTC) - Timestamp - date function – time function – Generating custom timestamp.

MODULE – III: Database handling using PHP and MySQL

Introduction to session – Inter page Communication - How session works on LAMP Server – Applications of session – Session management in PHP – drawbacks - Introduction to Cookies – need for cookie –Working principle – Implementation in PHP – Advantages of cookie over Session - Managing MySQL database with PHP – Establishing connection – Creating Query Strings – Executing Queries and Non-Queries – Fetching Data from result – Introduction to form Management – GET and POST forms – Retrieving data from forms – Repopulating HTML forms – Uploading files into server – validating uploaded files.

MODULE – IV: Client side scripting using javaScript

Introduction to client-side scripting - Need for a client-side language – Introduction to javaScript – History and evolution – Interpreter used for javaScript - javaScript basics - syntax and semantics – variables and scope – data types – operators – selection statements – loops – functions – strings and string operations – embedding javaScript in WebPages – arrays in JavaScript– HTML events and javaScript function - element selector functions – accessing html elements – modifying html elements – handling html events using event attribute function calls – accessing form elements – validating html forms.

TEXT BOOK(S)

1. HTML, CSS and JavaScript All in One, Sams Teach Yourself: Covering HTML5, CSS3, and jQuery Sams Publishing; 2 edition.

REFERENCE

1. PHP: The Complete Reference by Steven Holzner, McGraw Hill Education